



# *Caméléon Creator Tutorials*
# *Chapter 1*
*Create a work-flow prototype*

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***Caméléon Creator Tutorials***
***Chapter 1***
*Create a work-flow prototype*

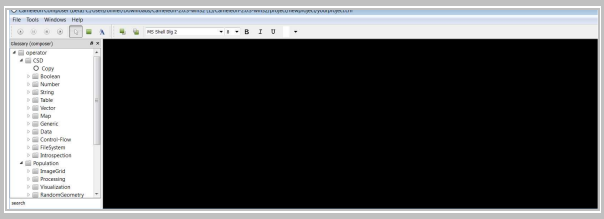
*By Olivier Cugnon de Sévricourt*

In this tutorial, we speak about how to use Caméléon to create a simple image processing prototype.

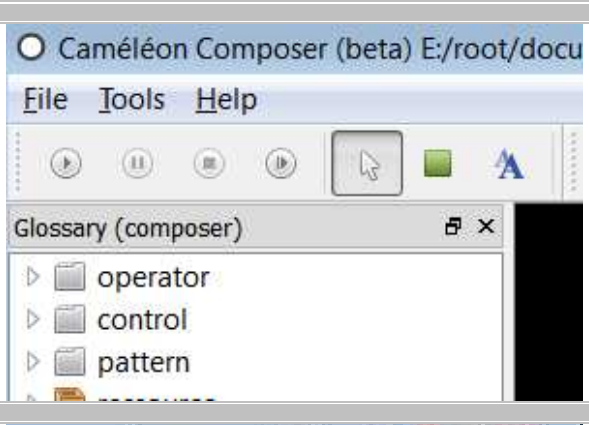
You will see all basics needed to **code graphically** using Caméléon.

You will also learn how to segment an image using a simple Threshold algorithm.

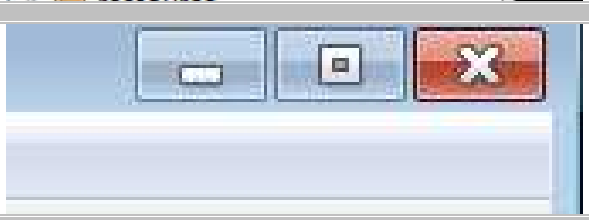
Segmentation is a way to partitioning the grey(color)-level image in term of segments (labeled image) as a prerequisite of the quantitative analysis.



**Create** a new project named *Threshold-Segmentation.cm*.



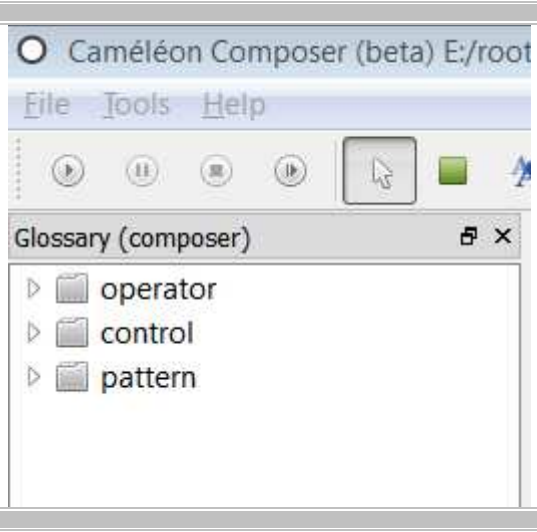

**Select** the composer window.



**Enlarge** it.

Warning: If you **close** the composer window, it **close** Caméléon creator.

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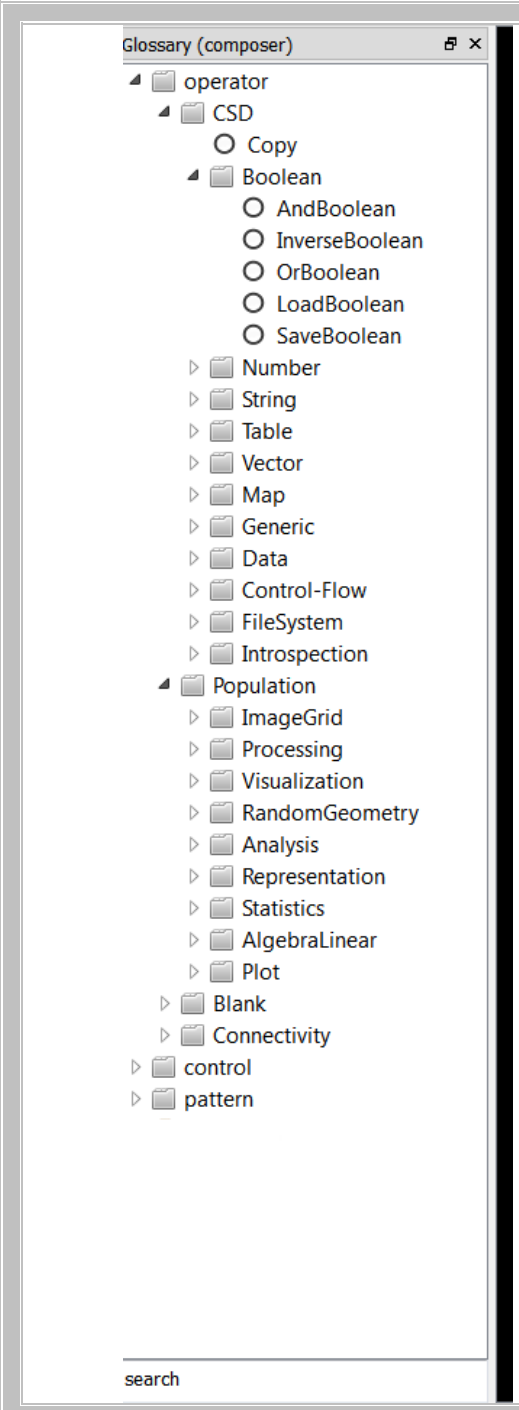

On the left side of the composer window, you can see the Glossary.

The glossary contains all elements you need to create prototypes & applications.

The glossary is divided in four parts.

For this tutorial, you focus on the two first parts: Operator & Control

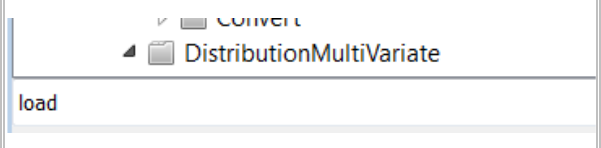
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If you **open** the Operator directory, you see many directories which contain directories or operators (with the circle icon representing a Caméléon Operator).

If you **open** the Control directory, you see many directories which contain directories or Controls (with the circle icon representing a Caméléon Control).

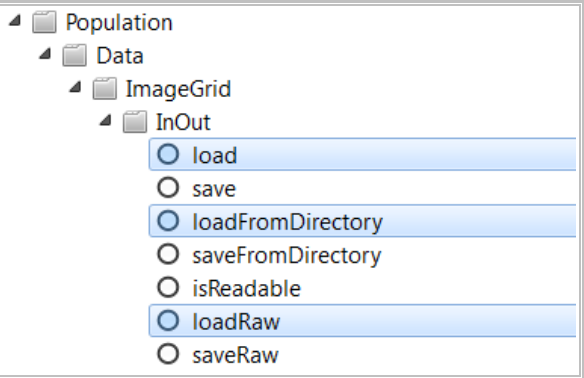
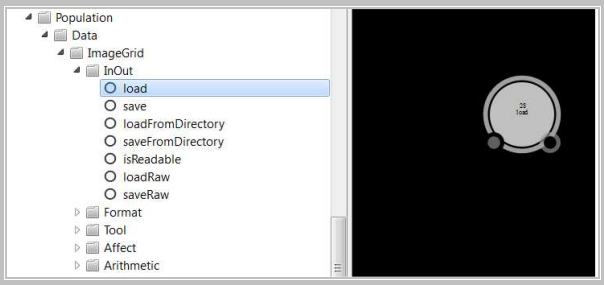
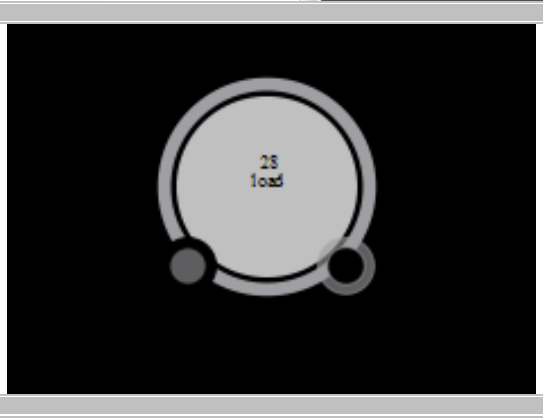
Now **search** some operators needed to build our segmentation prototype.



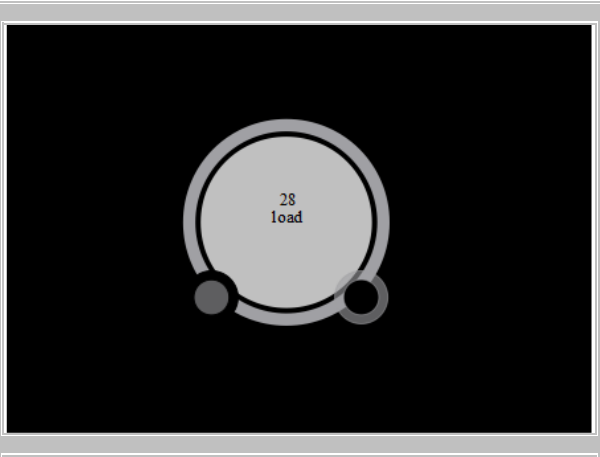
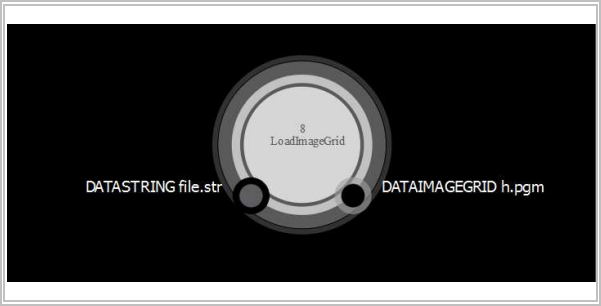
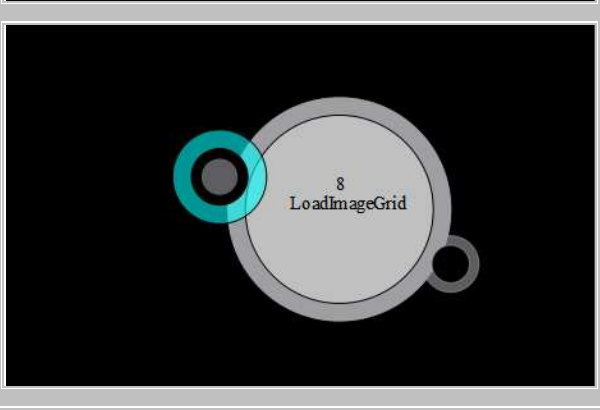

Start by using the **search** feature at the bottom of the glossary.

For this prototype we want to process an image, so let's **look** for a load image operator.

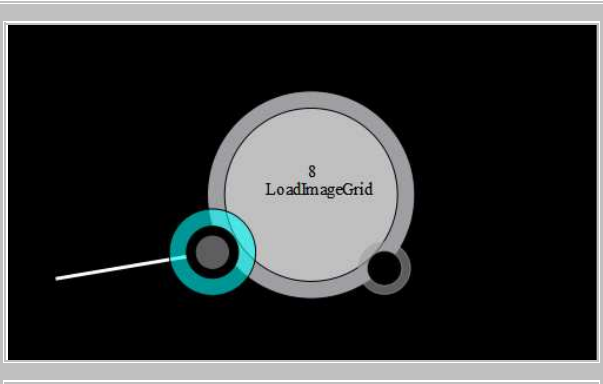
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| | |
|---|---|
| | **Write** load in the text box. |
|  | The search engine selects some opertors including one operator in operator > Population > Data > ImageGrid > InOut.<br><br>**Drag** the selected "load" name & **drop** it into the black space at the right of the glossary. |
|  | Great ! You have created your first operator ... |
|  | ... a Load ImageGrid, with one input (at the left) and one output (at the right). |
|  | If you **move** your mouse on top of this operator, an information tooltip appears.<br><br>This operator allows you to load an image from a file path. |

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| | |
|---|---|
|  | **Click** left on the operator to select it. |
|  | **Press** the W key on the keyboard to show operator information.<br><br>**Press** the W key again to hide information.<br><br>The input data is a string data type containing the file path & the output data is an image type containing the output image. |
|  | **Drag** the input connector to move it. |
|  | You can **change** the operator view by using this simple technics. |

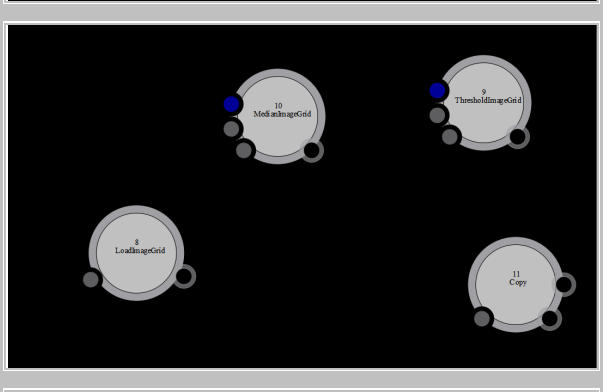
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If you **drag** the connector out of the operator circle you will be able to connect this operator to another.

But ... wait a minute ...

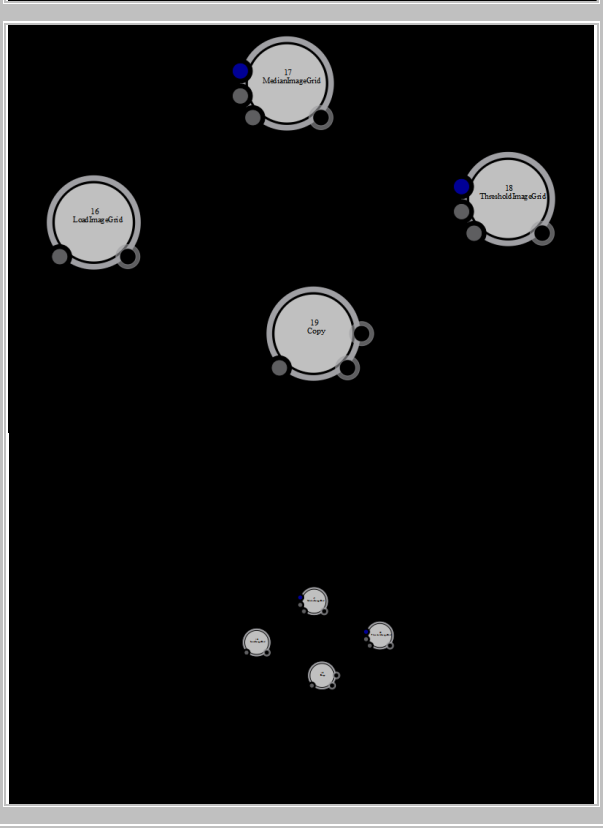
You don't have an other operator !!




So, let's **create** 3 operators:

Copy
Median
Threshold

**Find** them in the glossary and **create** them.



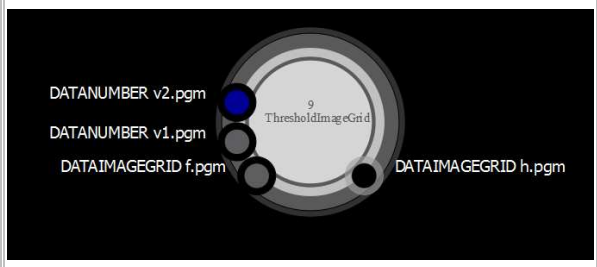
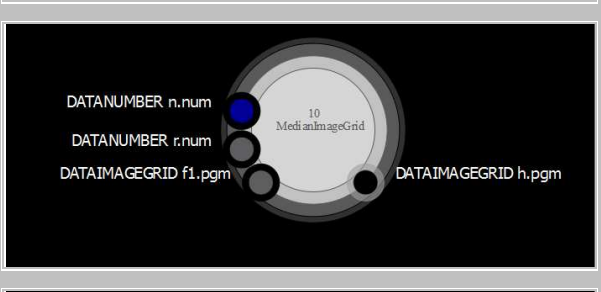
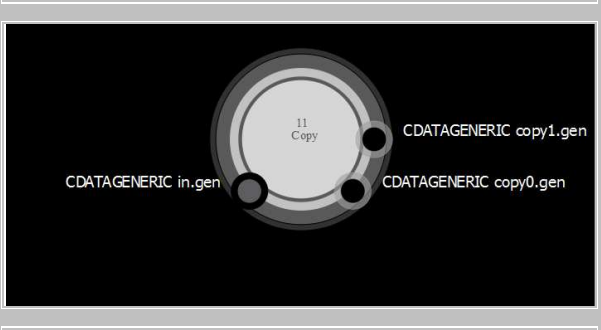
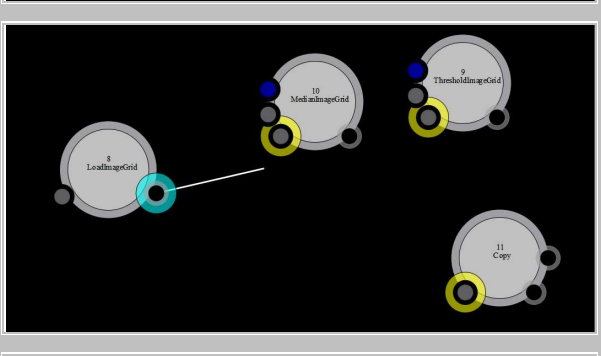
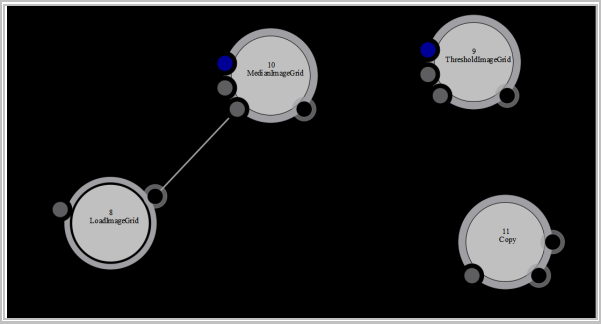

**Hold** the Q button to manage the program view.

**Hold** the Q button pressed, you can use drag & drop to change the view position.

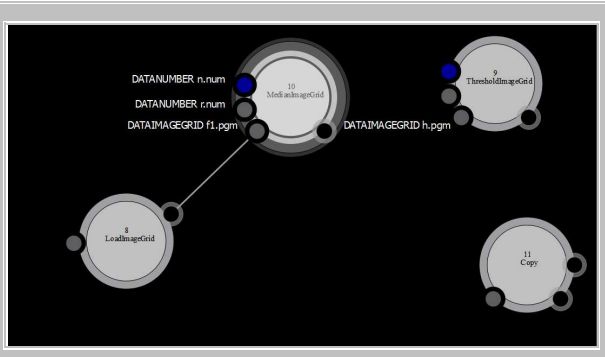
**Hold** the Q button pressed, you can use the mouse wheel to zoom in & zoom out.

You can also **use** + or - keyboard keys to **zoom in** & **zoom out**.

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| | |
|---|---|
|  | The Threshold is an operator which uses a grey level image & a threshold value to produce a binary image.<br><br>A white pixel is assigned to each pixel of the binary image which input image value is greater than the threshold value, a black pixel otherwise. |
|  | The Median<br>Operator smoothes image defects before segmentation. |
|  | The copy operator copies the input into two outputs.<br><br>By **clicking** right, and **selecting** action>add plug you **add** one more output. |
|  | If you **drag** the Load output connector out off the operator circle, you are able to **connect** this operator to another.<br><br>Yellow circles represent compatible connectors. |
|  | **Drop** the connection on the compatible input of Madian.<br><br>Great ! You have just **connected** one operator to an other ! |

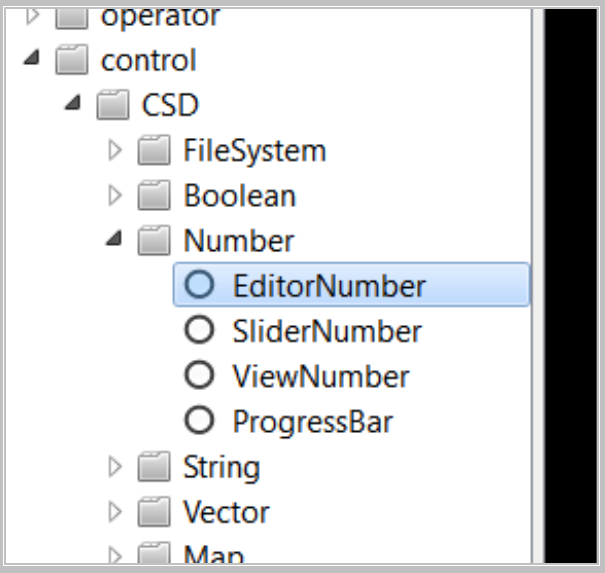
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**Show** Madian information by **pressing** the W key on the keyboard.

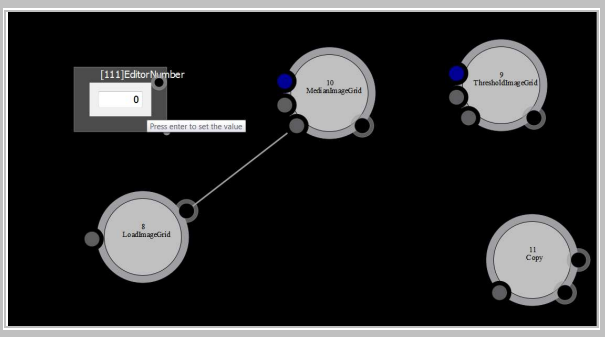
This operator needs an other input "r.num".

You **set** this value with a compatible "data control" with number.




Controls aims to visualize or edit data.

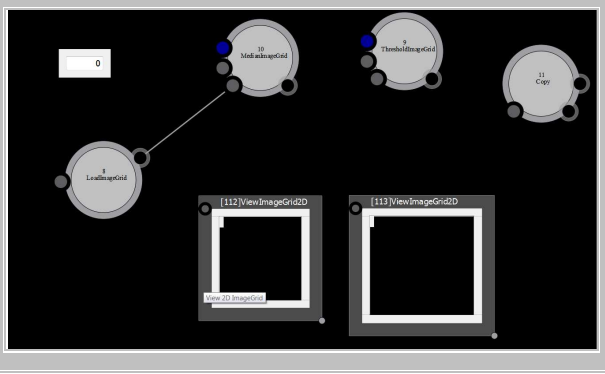
Now, **search** EditorNumber in the glossary.




**Create** an EditorNumber as you create an operator.

Like an operator, a control has input connectors at the left & output connectors at the right.

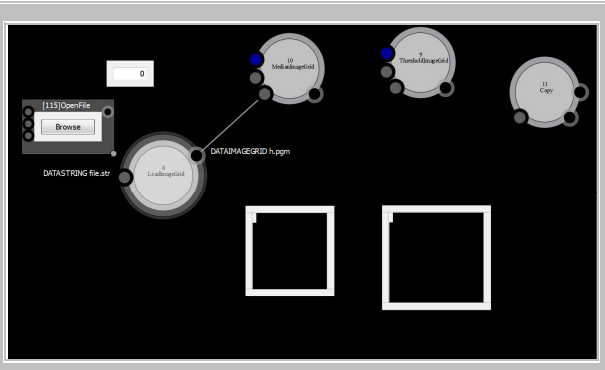
EditorNumber has just one output connector.




You need to **display** your images.

**Create** two ViewImageGrid2D.

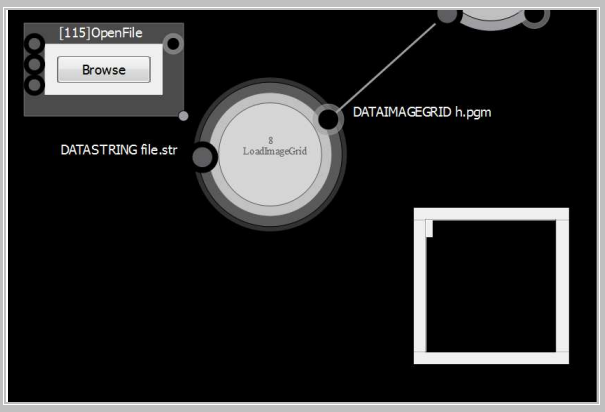
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You need to **set** the input image of our prototype.

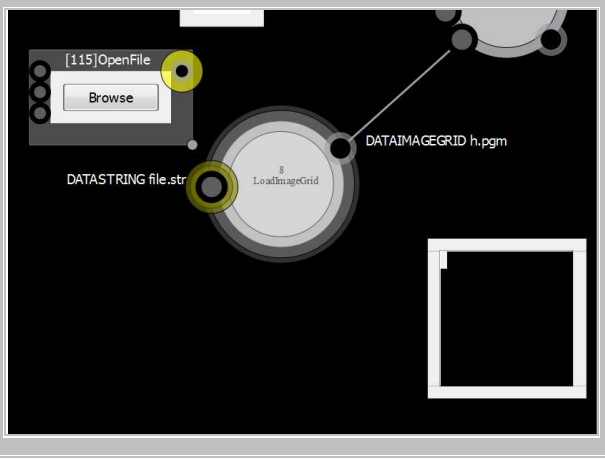
One way to do this is to use the OpenFile control.

Now, **create** one OpenFile control.




Now you want to **connect** the OpenFile output to the input of the Load in order to load our image into our process.

If you **hover** your mouse on the output connector, you see the data is the file path.

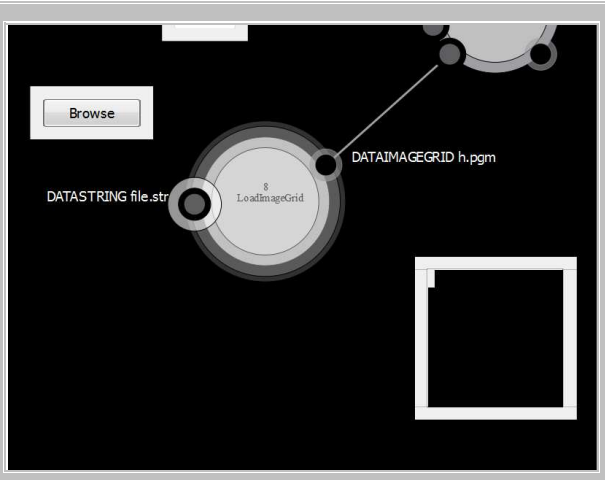
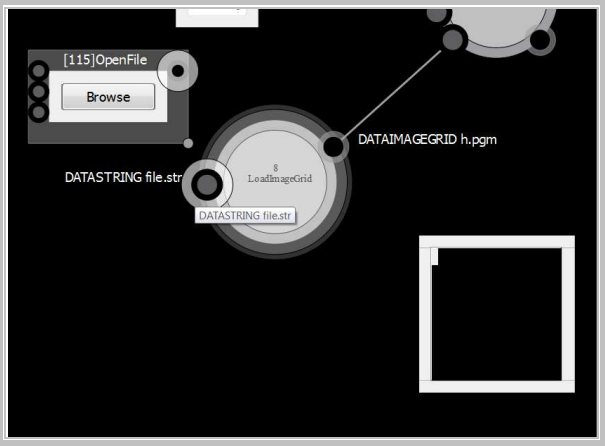
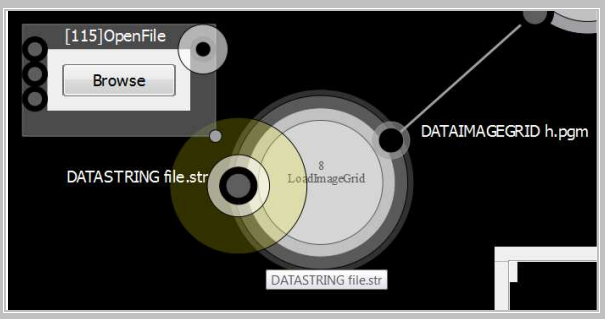
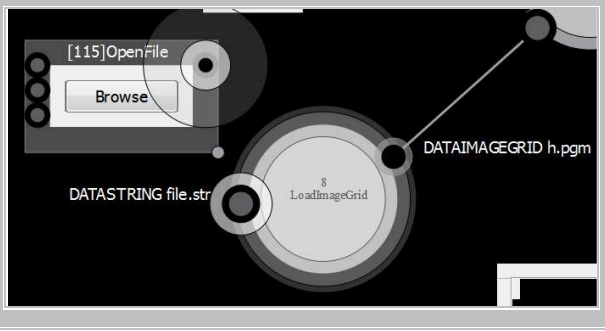



To start the connection mode, **click** one time on OpenFile output connector.

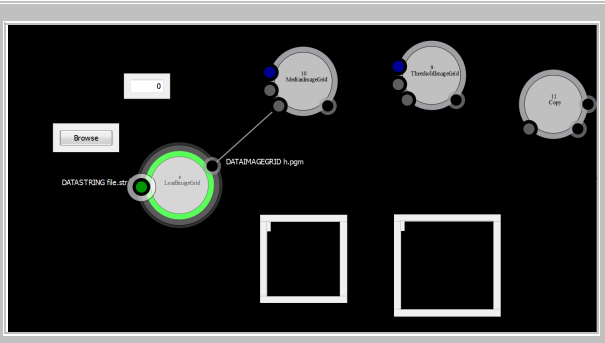
The connector show compatible connectors represented by a yellow circle around them. If you want to go out of this mode, you can **click** an other time on the OpenFile output connector.

In connection mode, **click** one time on the Load compatible connector to connect OpenFile & Load.

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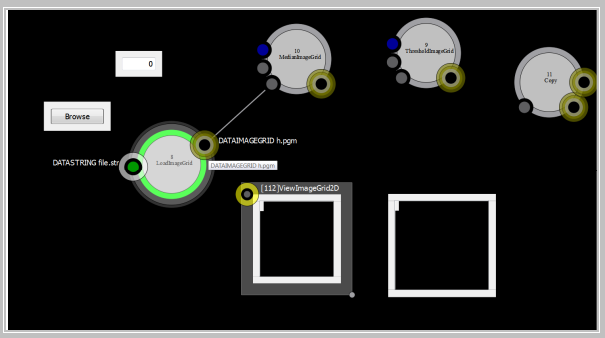
| | |
|---|---|
|  | A white circle is around Load input representing a connection with a control. |
|  | **Click** on OpenFile to see the output connector has the same state. |
|  | **Double click** on this connector to highlight the connected connector. |
|  | **Double click** on Load input to highlight the connected connector. |

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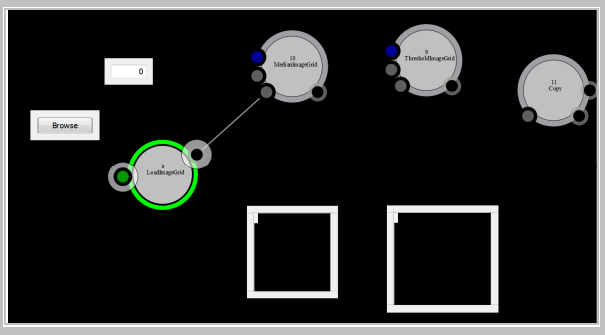



Now **click** on browse and find an image.

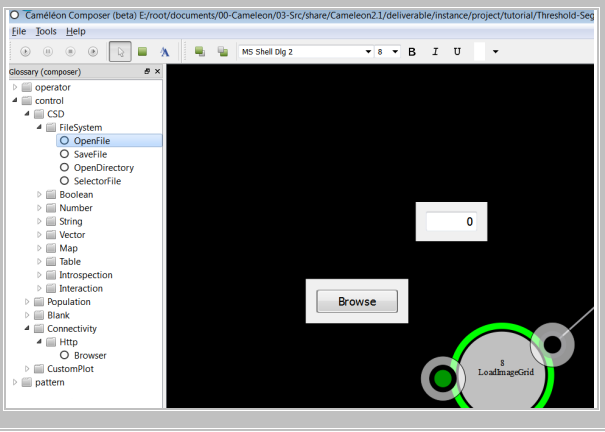
When you finish, your operator can be launched so it has a green colour.




**Connect** the first ViewImageGrid input to the Load load output.



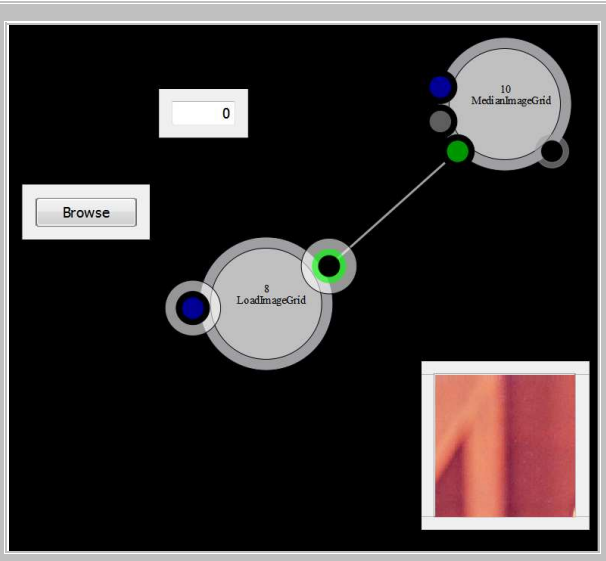

Like this.




At Composer top left, you can see a play button.

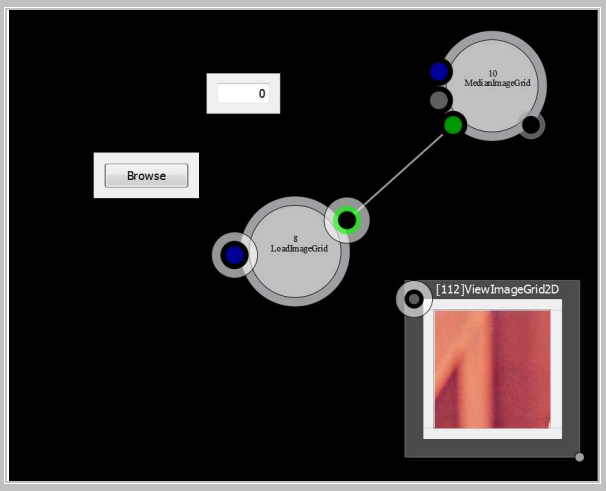
**Click** on this button to launch the Caméléon process mode.

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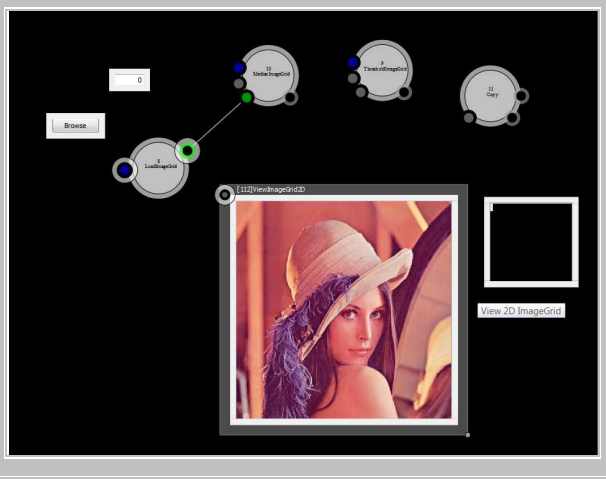



You can see your image in the view but the size is to small here.

Let's **select** the ViewImageGrid.




**Use** the bottom right grey circle to resize the view.

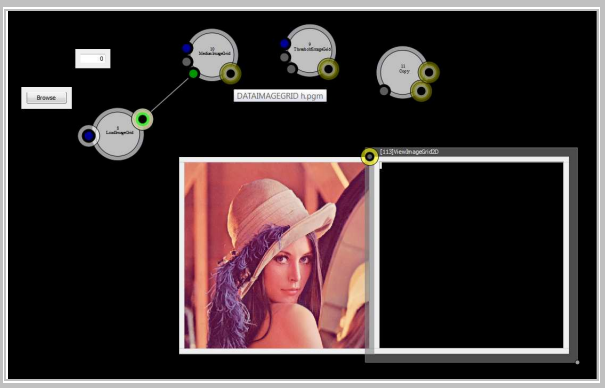



Like this.

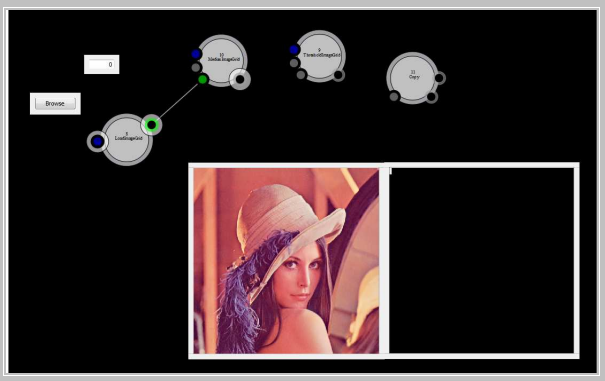
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Do the same for the second view.



**Connect** the second image view to the Median output.

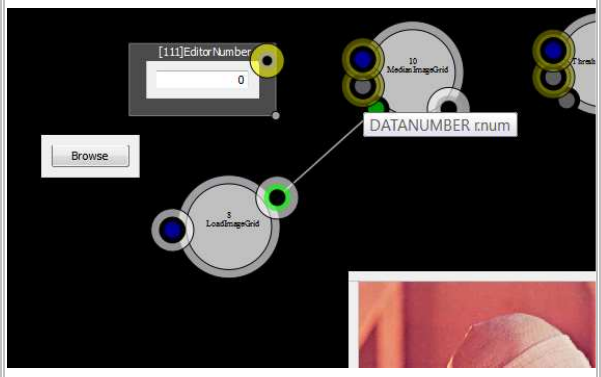


Like this.



Now **select** EditorNumber ...

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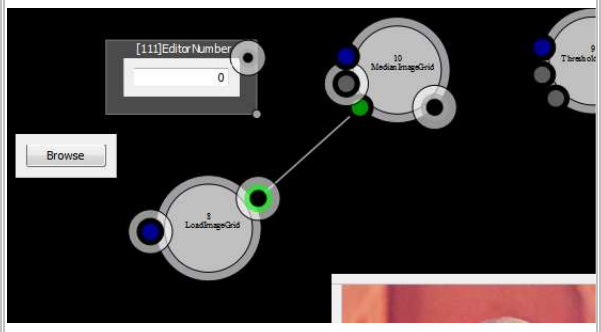

... and **select** the connector to connect it to Median.

Median has two unconnected inputs.

One has an inner grey colour, the other has an inner blue colour.

When a connector is blue and not connected, it's a connector with a default value.

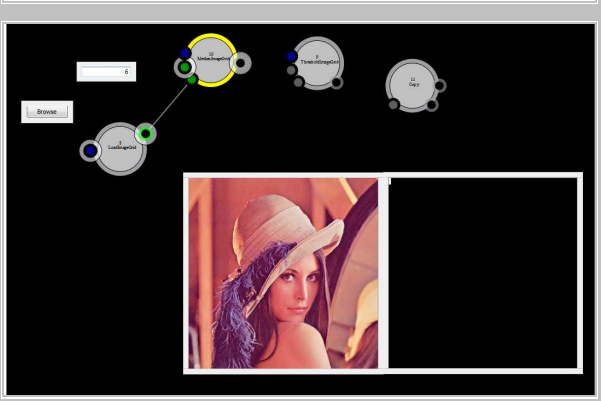
**Connect** EditorNumber to the connector with no default value.




Like this.

Now **set** a "6" value in the EditorNumber and **press enter** to send this value to the operator.

In edit mode, if you want to **correct** your value, **press** the backspace key on the keyboard. The DEL button is used for control & operator **deletion**.



The Caméléon virtual machine **executes** the Median operator represented by a yellow colour.

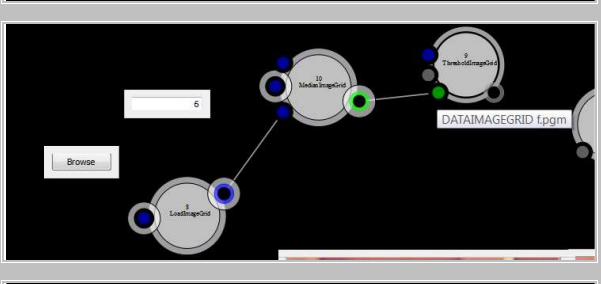
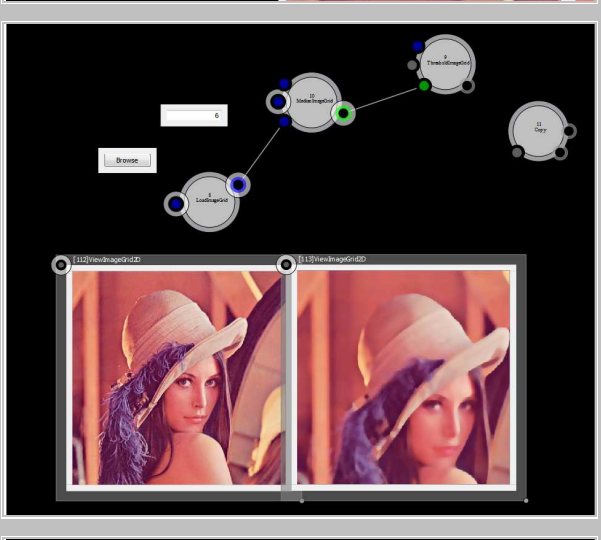
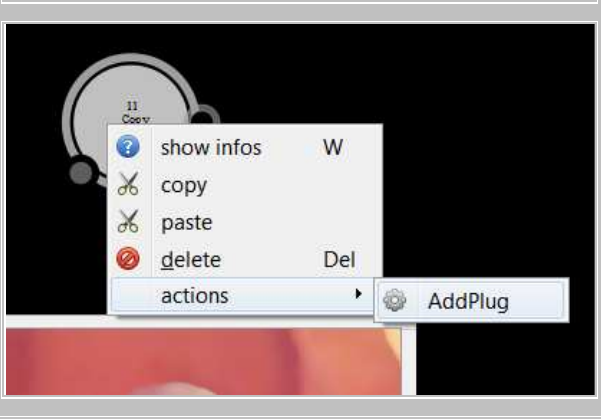


Now you can see the Median result on the right side.

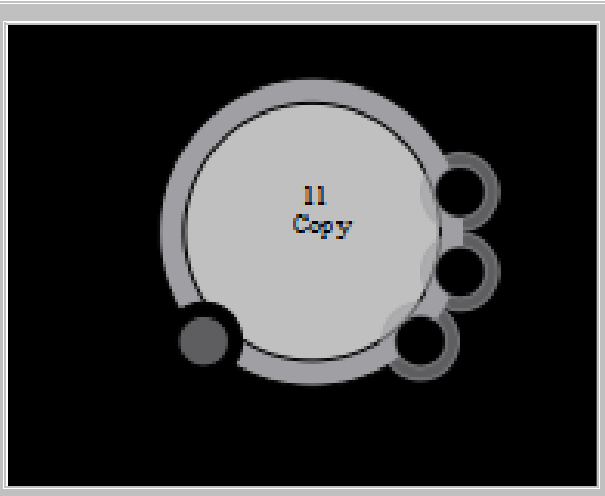
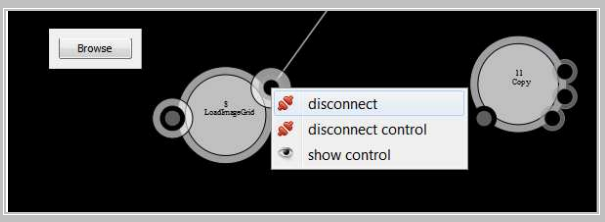
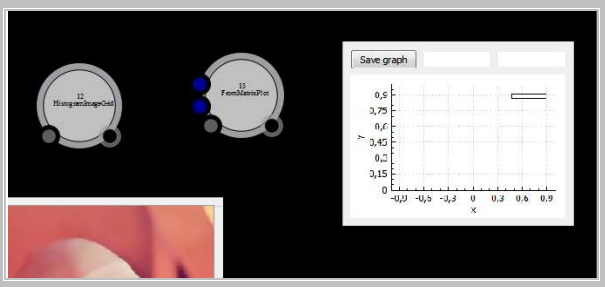
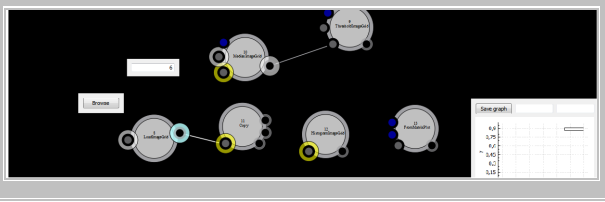
A connector with an inner blue colour contains an OLD data.

A connector with an inner or outer green colour contains a NEW data.

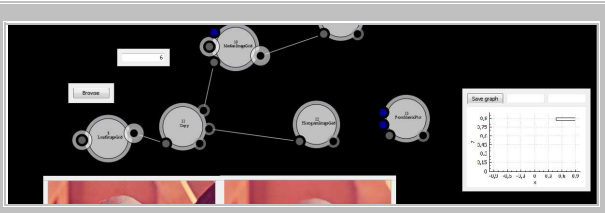
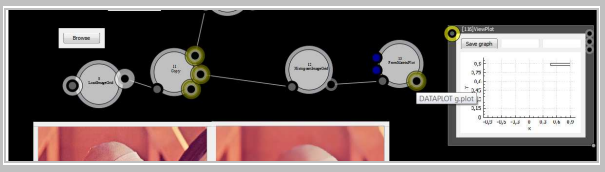
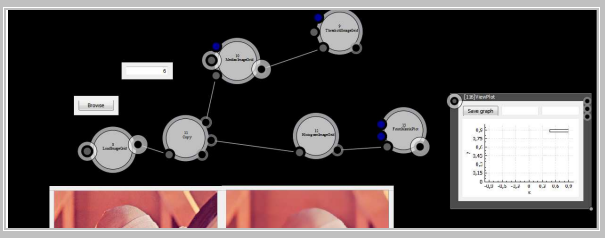
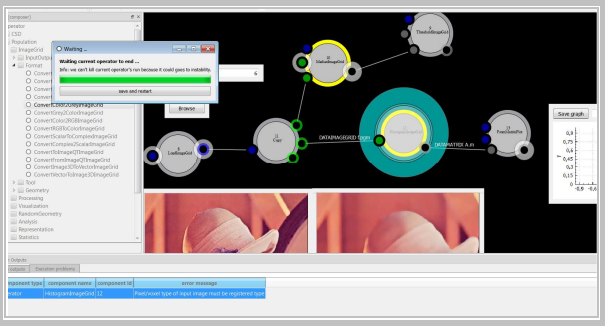
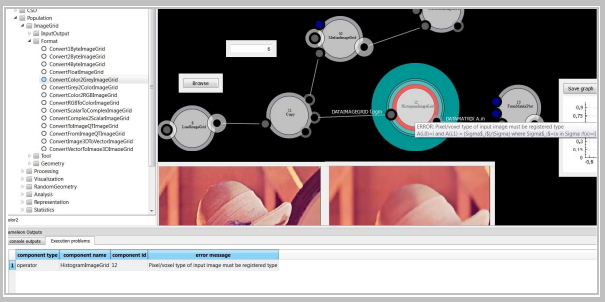
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| | |
|---|---|
|  | The result. |
|   | Now **connect** Median output to the Threshold input image. |
|   | **Create** a new copy operator. |
|   | **Add** a plug to this operator. |

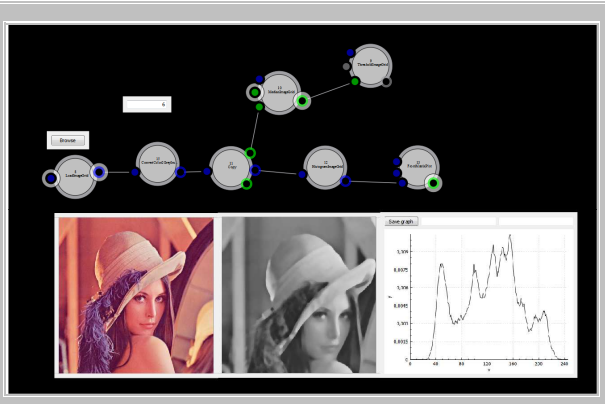
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| | |
|---|---|
|  | The copy operator can now make 3 data copies. |
|  | **Click right** on Load output connector, a context menu appear. **Press** on disconnect to **disconnect** the Load output. |
|  | **Create** two other operators:<br>Histogram<br>FromMatrix<br><br>And a control:<br>ViewPlot |
|  | Like this. |
|  | **Connect** load image output to copy input. |

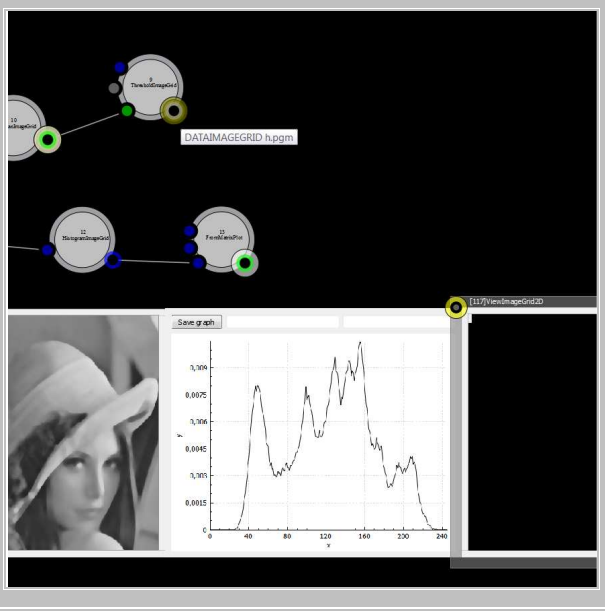
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| | |
|---|---|
|  | **Connect** one copy output to Medianinput and an other to Histogram input & **connect** Histogram output to FromMatrix non-default input. |
|  | **Connect** the ViewPlot input to FromMatrix output. |
|  | Like this.<br><br>**Run** the Machine. |
|  | Ooops !<br><br>An error occurs. |
|  | Histogram needs a grey level image. |
|  | **Create** a ConvertColour2Grey operator and connect it between Load & Copy (using **connect** & **disconnect**).<br><br>And **run** the machine. |

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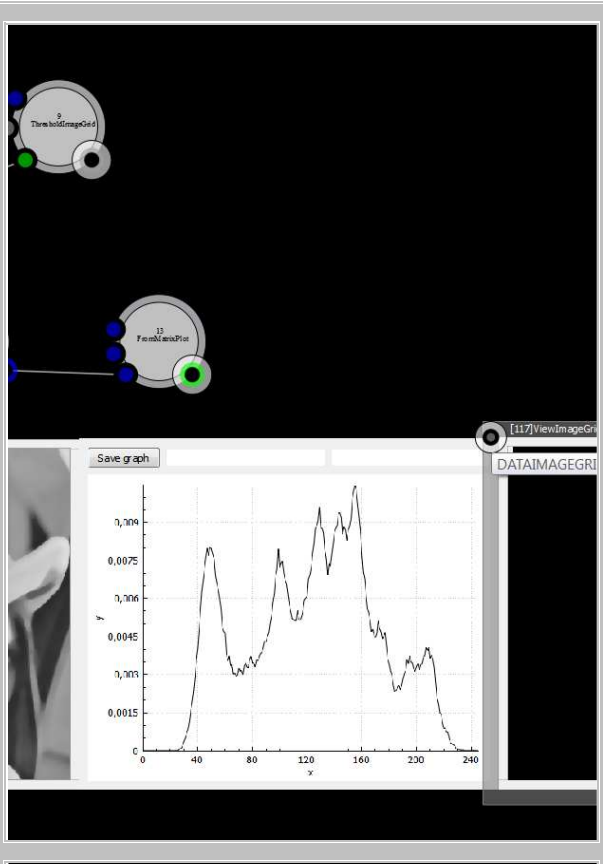


Like this.

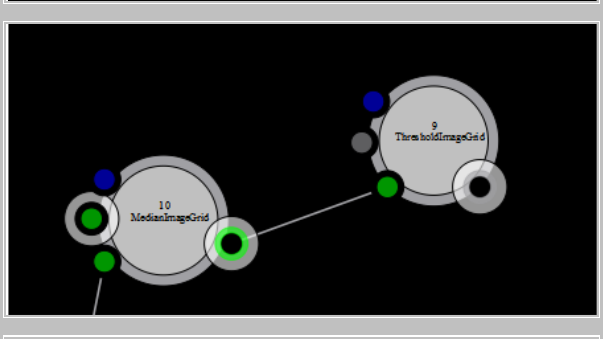



Let's **connect** an other image viewer at the Threshold output.

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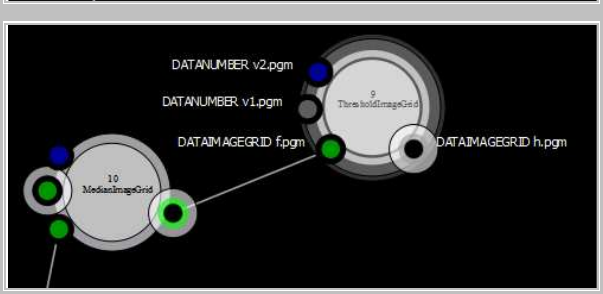



Like this.




Threshold has not been launched because one input connector does not contain a data.

You need to **set** this EMPTY data represented by an inner grey colour.

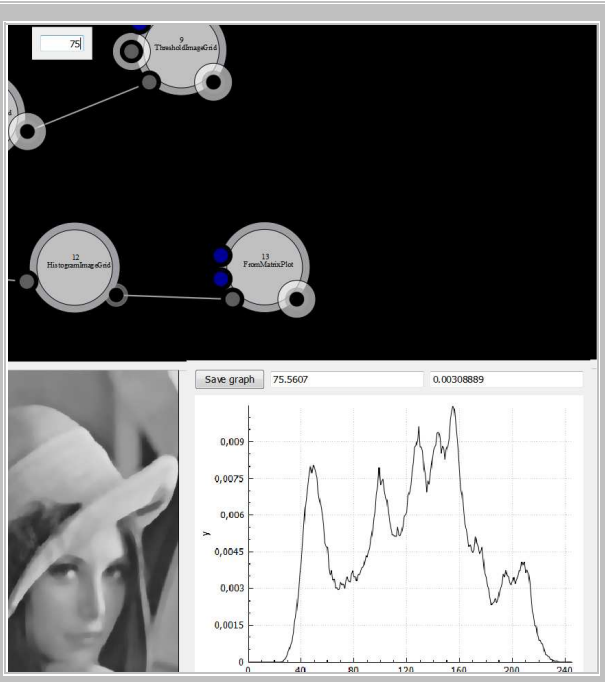



Let's see information about it.

You need an editor number.

**Create** it & **connect** it.

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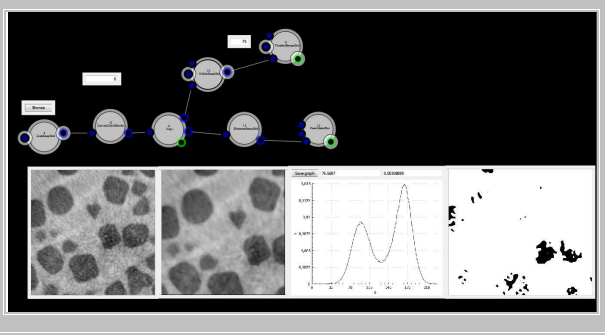

Like this (top left of the image).

**Set** the threshold value at 75 to the new EditorNumber and **press Enter**.



Like this.

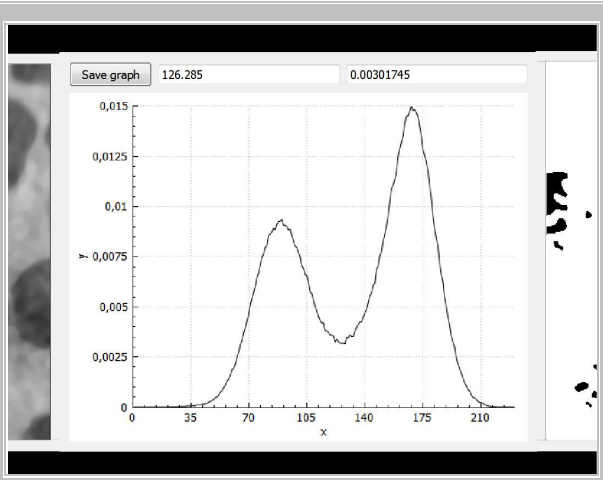
All pixel with a dark range are assigned with a black colour in the binary image & all the other are assigned with a white colour.



You **choose** an other image where this segmentation method works better.

The Threshold value is not well selected.

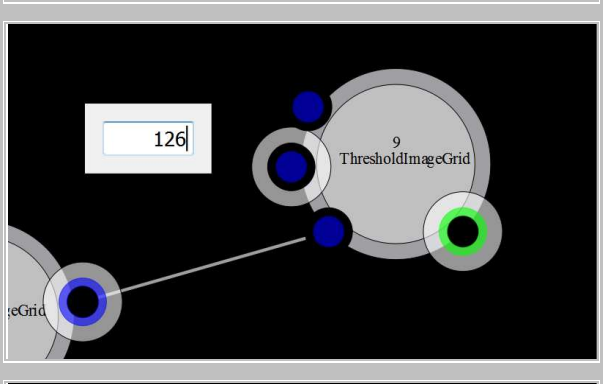
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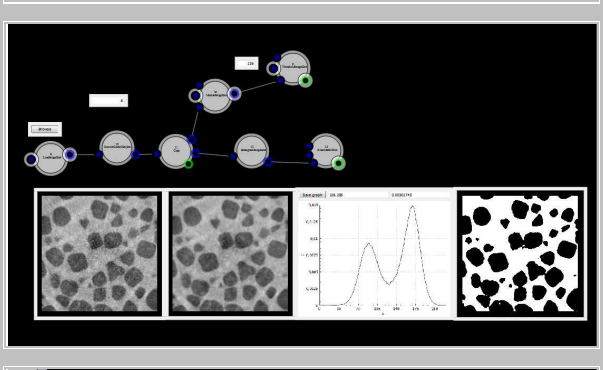

The two pics histogram means the image is populated by two pixel categories (dark grey at left, white grey at right).

To separate those, you **select** the point in the valley.

Approximately at x=126.

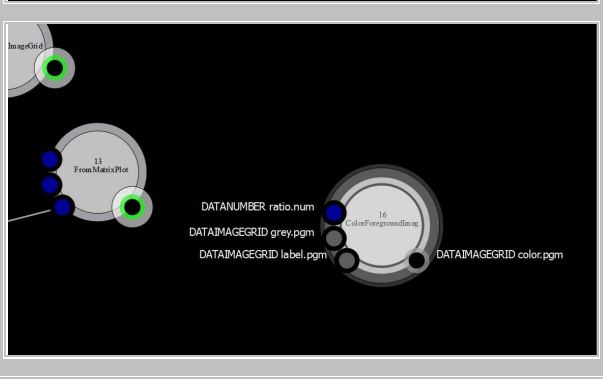



**Set** the Threshold value at 126.



Well, that's better.

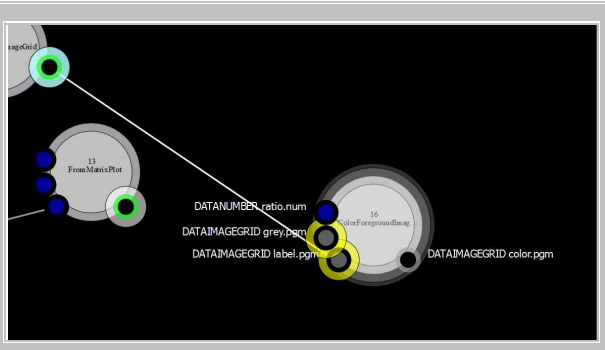
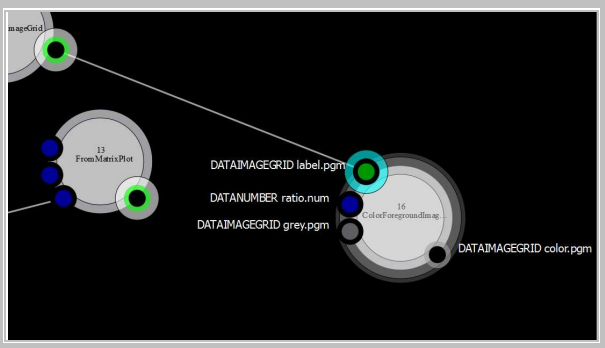
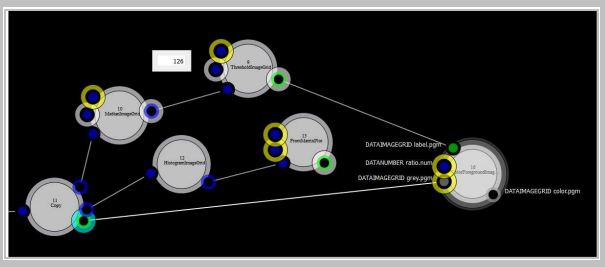
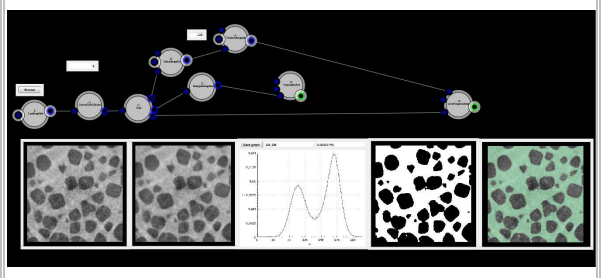
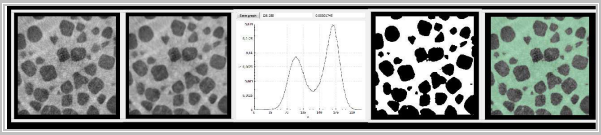
You now want to check that this segmentation is correct.



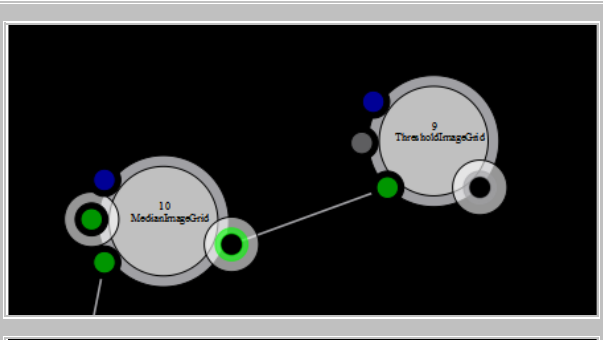

**Create** labelForeground operator.

This operator takes a grey scale image & set one colour to one label.

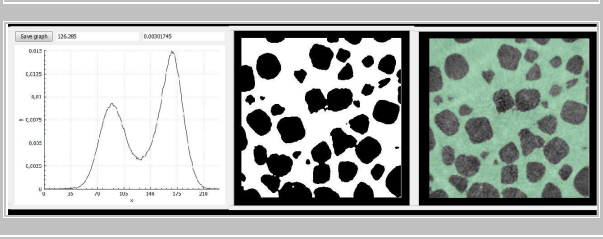
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| | |
|---|---|
|  | **Connect** the Threshold output to the label input of labelForeground. |
|  | **Move** the connector by a simple drag & drop. |
|  | **Connect** the last copy output to the grey input of labelForeground |
|  | **Save** your work by **pressing CTRL+S** or by using the **save feature in the file menu**.<br><br>Your project must be saved in an empty directory. So, you create a directory and save your file with a ".cm" extension in this directory.<br><br>In next tutorials, we refer to this project as *Threshold-Segmentation.cm*. |
|  | Nice ! |

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In this tutorial, we see how to **code graphically** using Caméléon creator & the Population dictionary.



The end.

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